

Technology Stack

Date	06 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit	Provides an interactive, responsive, and user-friendly web interface for recording/uploading audio, selecting topics, viewing evaluation results, and downloading reports with minimal development effort

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.11	Handles the core application logic, speech processing, semantic similarity evaluation, audio analysis, report generation, and integration with AI libraries efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	Local File Storage (CSV/JSON) & Session State	Stores user responses, evaluation results, and generated reports locally during execution. Suitable for the current standalone implementation and easy to extend with a database in future versions.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Streamlit Community Cloud (or Local Deployment)	Enables simple deployment and testing of the application while providing easy accessibility without complex server configuration

5	Version Control & CI/CD	Git & GitHub	Manages source code versions, tracks project progress, and maintains the project repository for collaboration, documentation, and deployment.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	OpenAI Whisper, Sentence-BERT, Librosa, PyDub, ReportLab, Pandas, NumPy	OpenAI Whisper converts speech to text, Sentence-BERT evaluates semantic similarity, Librosa and PyDub process audio features, ReportLab generates PDF reports, while Pandas and NumPy handle data processing and analysis